

# PL-300

## **Microsoft Power BI Data Analyst**

*Comprehensive study plan · From the fundamentals to certification*

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## 0. How to use this plan

This document covers five things: (1) a week-by-week study plan, (2) fundamentals explained in a teaching style, (3) concrete examples for each tool, (4) a deep dive into DAX and the Power BI interface, and (5) SQL/DAX best practices for optimizing performance. The interactive cheat sheet (HTML) is the companion quick-reference visual — use this document to learn in depth, and the cheat sheet to review.

### Comprehensive coverage

The exam does not forgive gaps just because a concept feels familiar from general BI experience: it asks for the exact name, the exact menu, and the exact behavior inside Power BI. That is why Section 9 exists — an exhaustive catalog of ALL visualization types, ALL filter types, and the rest of the ecosystem's functionality (Q&A, Copilot, dataflows, datamarts, embedded, paginated reports, etc.) — so no point of the official syllabus is left out of this plan.

### Where the real learning curve is

Most of the syllabus follows patterns common to any BI or data tool: tables, relationships, joins, dashboards. The steepest learning curve usually sits in three specific places: the M language (Power Query), the DAX engine (row context vs. filter context), and the Power BI Service administration ecosystem (workspaces, gateways, security). This plan is built to spend extra time exactly there.

Reference duration: 4 weeks at a pace of 10-12 hours per week (adjustable: compress it into 2 weeks by doubling daily hours, or stretch it to 8 weeks by slowing the pace — the topic order stays the same).

## 1. PL-300 exam overview

The exam certifies the “Power BI Data Analyst” role: someone capable of preparing data, modeling it, visualizing/analyzing it, and managing the Power BI environment securely. It consists of ~40-60 questions (multiple choice, drag-and-drop, case studies) in 100 minutes, with a passing score of 700/1000.

| Domain                            | Weight | What it essentially measures                                               |
|-----------------------------------|--------|----------------------------------------------------------------------------|
| 1. Prepare the data               | 25-30% | Power Query / M: connecting, cleaning, transforming and shaping the source |
| 2. Model the data                 | 25-30% | Star schema, relationships, DAX, model performance                         |
| 3. Visualize and analyze the data | 25-30% | Reports, storytelling, interactions, AI/Copilot, pattern detection         |
| 4. Manage and secure              | 15-20% | Workspaces, apps, gateways, RLS/OLS, content lifecycle                     |

Practical takeaway: the first three domains carry almost the same weight (~80% of the exam combined), so none of them can be neglected. The fourth domain is shorter but very “memorization-heavy” (roles, gateway types, menu options) — it is best studied last.

## 2. Week-by-week study plan

### Week 1 · Preparing data (Power Query / M) — Domain 1

Goal: be able to connect to any source, profile data quality, and shape it into a clean star schema, while understanding what preserves and what breaks query folding.

### Day 1-2: Installation, interface and first connection

- Install Power BI Desktop (free, Windows). On Mac, install a Windows VM or use Power BI Service in the browser for the reporting parts (Power Query Desktop has no native Mac version).
- Tour the interface: ribbon, Data pane, Fields pane, Visualizations pane, and the three views (Report / Data / Model) — see Section 4 of this document for the detailed tour.
- Connect to a simple source (Excel or a sample CSV) and open the Power Query Editor.

### Day 3-4: Data transformation

- Practice: changing data types, splitting/merging columns, conditional columns, grouping and aggregating (Group By), pivoting/unpivoting, transposing.
- Learn Merge (equivalent to a JOIN) vs Append (equivalent to a UNION).
- Practice handling import errors and null/inconsistent values; use the column profile (Column Quality/Distribution/Profile).

### Day 5: Source modeling and query folding

- Design one fact table and several dimension tables from a denormalized source (a classic exam exercise).
- Understand which steps DO fold back to SQL at the source (query folding) and which ones break it: custom indexes, certain merges, non-foldable M functions, combining queries from different sources.
- Configure at least one parameter (for example, to point to Dev/Prod, or to drive an incremental refresh).

#### Concrete example

Connect to a SQL view (not the raw table), apply a date filter, and check under “View Native Query” that Power Query is still generating SQL (folding active). Then add a custom index column and watch that option disappear: you have just broken folding.

### Day 6-7: Review and lab

- Compare DirectQuery vs Import vs Dual using a real table: measure the time it takes to change a date on a slicer under each mode.
- Domain 1 review quiz (use the cheat sheet, Data Preparation section).

## Week 2 • Modeling data and DAX — Domain 2

Goal: build a solid star-schema model and write correct DAX measures using CALCULATE, time-intelligence functions and row-level security.

### Day 1-2: Star schema and relationships

- Set up 1:N relationships with single filter direction (avoid bidirectional unless there is a justified case).

- Create your own date table with CALENDAR/CALENDARAUTO and mark it as a date table (Mark as Date Table).
- Implement a role-playing dimension, e.g. a Date table used for both Order Date and Ship Date via inactive relationships + USERRELATIONSHIP.

#### **Day 3-4: DAX fundamentals**

- Row context vs. filter context (the single most-tested concept on the exam).
- CALCULATE as the context-manipulation engine: ALL, ALLEXCEPT, ALLSELECTED, KEEPFILTERS.
- Simple aggregation measures, basic statistical functions (AVERAGE, MEDIAN, STDEV.P, RANKX).

#### **Day 5: Time intelligence and semi-additive measures**

- DATEADD, SAMEPERIODLASTYEAR, DATESYTD/QTDMTD, TOTALYTD, PARALLELPERIOD.
- Semi-additive measures (e.g. inventory balance with LASTNONBLANK / LASTDATE).
- Calculated columns and calculated tables: when to use them (rarely) instead of measures.

#### **Day 6: Calculation groups and quick measures**

- Create a calculation group so you don't need to duplicate 5 measures x 3 time variants (via Tabular Editor or the native Desktop feature).
- Generate a Quick Measure and then “edit” it to see the DAX Power BI produced — an excellent way to learn patterns.

#### **Day 7: Model optimization**

- Use Performance Analyzer to measure which visual/measure is slow.
- Remove unnecessary columns/rows, reduce granularity, disable “Auto Date/Time”.
- Domain 2 review (using the cheat sheet, DAX + Modeling section).

### **Week 3 · Visualization, analysis and storytelling — Domain 3**

Goal: build usable, accessible, analytically rich reports, and master the AI/Copilot features now covered by the exam.

#### **Day 1-2: Visuals and formatting**

- When to use each visual (bars vs. lines vs. cards vs. matrix vs. map) — the criterion is the business question, not aesthetics.
- Conditional formatting, custom themes (theme JSON), report page setup.

#### **Day 3: Interactivity**

- Bookmarks, editing interactions between visuals (filter/highlight/none), custom tooltips.
- Drillthrough (pages, filters, buttons) and page navigation.
- Sync Slicers across pages.

#### **Day 4: Advanced analytics and AI**

- Decomposition Tree, Key Influencers, trend lines, error bands, forecasting.
- Anomaly/outlier detection, grouping (grouping/binning/clustering).
- Copilot: creating a new page, suggesting content, generating a narrative visual, summarizing the semantic model.
- Visual calculations — the most recent addition to the syllabus.

#### **Day 5: Accessibility, export and mobile**

- Tab order, alt text, theme contrast, mobile layout design.
- Export settings, page auto-refresh, end-user report personalization.

#### **Day 6-7: Integrative lab and review**

- Build a 3-page report (executive summary, detail, drillthrough) with at least: 1 bookmark, 1 custom tooltip, 1 drillthrough, and one synced slicer.
- Domain 3 review using the cheat sheet.

### **Week 4 · Management, security and final review — Domain 4**

Goal: understand the content lifecycle in the Power BI Service (workspaces, apps, gateways, RLS) and arrive at the exam having completed practice tests.

#### **Day 1-2: Workspaces and distribution**

- Workspace roles (Viewer, Contributor, Member, Admin) and what each one can do.
- Power BI apps, dashboards vs. reports, distribution methods, subscriptions and data alerts.
- Certifying/promoting content, deployment pipelines (Dev/Test/Prod).

#### **Day 3: Gateways and refresh**

- Standard vs. personal gateway, and when a gateway is NOT needed (native cloud sources vs. on-premises).
- Configuring scheduled refresh for a semantic model.

#### **Day 4-5: Security (RLS/OLS) and governance**

- Static and dynamic RLS roles with USERPRINCIPALNAME()/USERNAME(), RLS group membership.
- Object-level security (OLS), item-level access, semantic model access, sensitivity labels.

#### **Day 6-7: Mock exams and cross-cutting review**

- Take at least 2 full practice exams under real time conditions (100 minutes).
- Review every mistake by going back to the exact concept in this document or the cheat sheet — don't memorize the question, understand the reasoning.
- Final review: read through the entire cheat sheet in one sitting, out loud if possible.

## 3. Detailed explanation by domain

### 3.1 Preparing the data

#### Data sources and connectivity modes

Power BI supports three data-storage strategies within the model:

| Mode        | What it does                                                                                   | When to use it                                                                                      |
|-------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Import      | Copies the data and compresses it in the VertiPaq engine (in memory)                           | Most cases: maximum performance, no permanent connection required                                   |
| DirectQuery | Every interaction sends a query to the source; no rows are stored                              | Near real-time data, volumes too large to fit in memory, data-governance requirements at the source |
| DirectLake  | Reads Delta/Parquet files from OneLake (Fabric) directly, without copying or using DirectQuery | Fabric/OneLake with large volumes: combines Import's speed with DirectQuery's freshness             |

A “composite” model combines several sources with different modes (e.g. one table in DirectQuery and the rest in Import), or uses “Dual” mode on a dimension so it behaves as Import when combined with Import tables and as DirectQuery when combined with DirectQuery tables.

#### Profiling and cleaning

- Column quality / Column distribution / Column profile (on the “View” ribbon of the Power Query editor) show the % of errors, blanks, and distinct values — always check these before transforming blindly.
- Import errors usually come from badly inferred data types: set the type explicitly in the first step instead of relying on auto-detection.

#### Transform and load

- Fact tables: contain the numeric metrics and the foreign keys pointing to the dimensions. Make them as “long” (many rows) and “narrow” (few columns) as possible.
- Dimension tables: contain descriptive attributes (text, categories, dates) used for filtering and grouping.
- **Reference vs Duplicate:** a “Referenced” query reuses the steps of the original without copying them (changes at the source propagate); “Duplicate” creates an independent copy. Use a reference when you want to derive several clean tables (facts + dimensions) from a single staging query.
- **Merge (JOIN):** combines columns from two queries by a key, with the same join types as SQL (Left Outer, Inner, Right Outer, Full Outer, Anti).
- **Append (UNION):** stacks rows from two or more queries with a compatible column structure.

```
// M example: profiling + type + conditional column
let
    Source = Sql.Database("server", "Sales_DB"),
    Sales = Source{[Schema="dbo",Item="Sales"]}[Data],
```

```

FixedTypes = Table.TransformColumnTypes(Sales, {"Date", type date}, {"Amount", type
number})),
Segment = Table.AddColumn(FixedTypes, "Segment",
    each if [Amount] > 1000 then "High" else "Standard")
in
    Segment

```

### Exam-critical

If asked “which step breaks query folding”, suspect first: Table.AddIndexColumn, Table.Buffer, combining different sources, or steps that use M functions with no SQL equivalent.

## 3.2 Modeling the data

### Star schema

The star schema is the design Power BI expects: a central fact table surrounded by conformed dimensions, all with 1:N relationships and “Single” filter direction. Avoid a snowflake schema unless truly necessary, and avoid direct many-to-many relationships (use a bridge table if unavoidable).

### DAX: the two contexts

- **Row context:** exists when DAX evaluates row by row (calculated columns, or inside iterators like SUMX/FILTER). There you can reference Table[Column] of that row directly.
- **Filter context:** the set of active filters (from slicers, matrix rows/columns, or CALCULATE) that determines which rows get aggregated in a measure.
- **Context transition:** happens when CALCULATE (or an iterator) converts a row context into an equivalent filter context — this is the concept that trips up newcomers the most, and the most heavily tested piece of the exam.

```

// Basic measure
Total Sales = SUM ( Sales[Amount] )

// CALCULATE modifies the filter context
Prior Year Sales =
CALCULATE ( [Total Sales], SAMEPERIODLASTYEAR ( Calendar[Date] ) )

// ALLEXCEPT: removes all filters from the table except the ones listed
% of Category Total =
DIVIDE (
    [Total Sales],
    CALCULATE ( [Total Sales], ALLEXCEPT ( Product, Product[Category] ) )
)

// Variables to avoid recalculating
Margin % =
VAR Revenue = [Total Sales]
VAR Cost = [Total Cost]
RETURN DIVIDE ( Revenue - Cost, Revenue )

```

### Time intelligence and semi-additive measures

- DATESYTD/QTD/MTD, SAMEPERIODLASTYEAR, DATEADD, PARALLELPERIOD all require a contiguous date table marked as a “Date table”.
- Semi-additive measures (balances, stock) don't sum over time: take the last available value per period with LASTDATE/LASTNONBLANK.

### Calculation groups

A calculation group lets you apply the same “lens” (e.g. Actual / YTD / Prior Year / Var%) to any base measure without duplicating DAX code. Create them from Power BI Desktop (Modeling > New calculation group) or with Tabular Editor for finer control.

### Model optimization

- Remove unused high-cardinality columns (IDs, millisecond timestamps).
- Split DateTime into Date + Time when you don't need both granularities together.
- Reduce granularity (aggregate at the source or in Power Query) when row-level detail isn't needed.
- Use Performance Analyzer and the DAX query view (or DAX Studio) to isolate slow measures/visuals and inspect the query plan.

## 3.3 Visualizing and analyzing the data

This domain tests design judgment (choosing the right visual), interaction mechanics (bookmarks, drillthrough, tooltips), and the analytical/AI features Power BI layers on top of standard visuals.

| Business need                       | Recommended visual                         |
|-------------------------------------|--------------------------------------------|
| Compare categories                  | Bar/column chart                           |
| See a trend over time               | Line / area chart                          |
| Composition of a total              | Pie chart (few categories) or stacked bars |
| Tabular detail with totals          | Matrix, not a plain table                  |
| Relationship between two measures   | Scatter chart                              |
| A single highlighted KPI            | Card / KPI visual                          |
| Explore the “why” behind a variance | Decomposition tree / Key Influencers       |

- **Bookmarks:** capture the filter/visibility state of a page; combined with buttons, they simulate tabs or tutorial steps.
- **Drillthrough:** navigates to a detail page, passing along the filter context of the row/point clicked; configured by dragging a field into the “Drillthrough” well of the destination page.
- **Copilot:** generates new pages, suggests visuals, drafts a narrative summary of the report, and summarizes the semantic model in natural language — part of the current syllabus, worth reviewing even without a Premium license to try it firsthand.

- **Visual calculations:** DAX defined directly on top of a visual's already-aggregated data, useful for rankings or differences between the visual's own columns without touching the model.

### 3.4 Managing and securing Power BI

| Workspace role | Can                                                                                |
|----------------|------------------------------------------------------------------------------------|
| Viewer         | View published content; cannot edit                                                |
| Contributor    | Publish/edit reports and models, without managing the workspace                    |
| Member         | Everything Contributor can do, plus add/manage other users (depending on settings) |
| Admin          | Full control: delete the workspace, manage all roles and settings                  |

- **Standard gateway:** installed on an on-premises server, shared, supports scheduled refresh and DirectQuery against local systems.
- **Personal gateway:** single-user only, does not support DirectQuery in most cases, intended for one PC.
- **No gateway needed:** when the source is already natively reachable over the internet by the Power BI Service (e.g. a cloud-hosted database with a public endpoint and cloud credentials).
- **Static RLS:** a fixed filter per role, e.g. Role="Spain" → Table[Country]="Spain".
- **Dynamic RLS:** the filter depends on who opens the report, using USERPRINCIPALNAME() or USERNAME() cross-referenced against a user→region mapping table.

```
// Typical dynamic RLS rule
[User Email] = USERPRINCIPALNAME ()
// combined with a Users[Email] -> Users[Region] table
// and a relationship Users[Region] -> Dim_Region[Region]
```

## 4. Power BI Desktop: interface tour

If you have never opened Power BI Desktop before, the fastest way to get oriented is to separate its three layers: the interface for connecting and shaping data (Power Query), the semantic layer where relationships and DAX measures live (the Model), and the canvas where visuals are assembled (the Report).

| View in Power BI Desktop | What it's for                                                |
|--------------------------|--------------------------------------------------------------|
| Report view              | Design pages, drag visuals onto the canvas, apply formatting |
| Data view                | Browse the loaded rows, create quick columns/measures        |
| Model view               | Define relationships, cardinality, and filter direction      |
| Power Query Editor       | Connect to, clean, and transform data before it is loaded    |

- **Fields pane:** lists the tables and columns in the model.

- **Visualizations pane:** a gallery of visual types plus “wells” where you drag fields onto Axis, Legend, Values, etc.
- **Filters pane:** filters at the visual, page, or report level, with a visible hierarchy between them.
- **Ribbon:** Home, Insert, Modeling, View, Optimize (Performance Analyzer lives here), Help.

#### Recommended first exercise

Open Power BI Desktop, connect to a sample Excel file, go to Model view, and mentally sketch out which table would be the “fact” and which would be “dimensions” before creating any relationship. That mental habit accounts for roughly half of Domain 2 on the exam.

## 5. Other applications in the Power BI ecosystem

| Tool                               | What it is                                                                      | When you need it for the PL-300                                      |
|------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Power BI Desktop                   | Authoring app (Windows): Power Query + model + reports                          | 80% of your practice happens here                                    |
| Power BI Service (app.powerbi.com) | Web platform: workspaces, apps, gateways, subscriptions                         | All of Domain 4 is studied and practiced here                        |
| Power Query Online                 | M editor inside the Service / Dataflows                                         | Dataflows shared across several models                               |
| DAX Studio                         | Free external tool for running/tuning DAX queries and inspecting the query plan | Advanced measure optimization (appears in the performance syllabus)  |
| Tabular Editor                     | External editor for the tabular model (BIM)                                     | Creating calculation groups and bulk edits faster than the native UI |
| On-premises Data Gateway           | Secure bridge between local sources and the Service                             | Refreshing/querying data not exposed to the internet                 |
| Paginated Reports (Report Builder) | Fixed-layout reports, invoice/PDF style                                         | Cases where the user needs an exact print/export layout              |

## 6. SQL, Power Query and DAX best practices for performance

This section organizes optimization guidance by layer: first the source database, then the load stage (Power Query), and finally the model/DAX.

### 6.1 At the SQL source

- **Views, not raw tables:** expose SQL views to Power BI with only the necessary columns and already-corrected types; don't make Power Query “guess” types.
- **Strategic indexes:** create B-Tree indexes on foreign keys used as relationships, and composite or date-partitioned indexes on columns used as temporal filters.
- **Push computation to the source:** heavy string/format calculations should be resolved in materialized columns or in the view, not row-by-row in Power Query.

- **Respect query folding:** avoid custom SQL embedded in the connector if it breaks downstream folding; encapsulate that logic in a view at the source instead.

## 6.2 In Power Query

- Filter and select columns as early as possible in the step sequence (reduce volume as soon as you can).
- Check “View Native Query” after each critical step to confirm folding is still active.
- Use Table.Buffer only when explicitly needed (it intentionally breaks folding, e.g. to stabilize an unstable source) — never by default.
- Configure incremental refresh (RangeStart/RangeEnd) on large fact tables (5M+ rows) that don't need real-time freshness.

## 6.3 In the model and DAX

- **Reduce cardinality:** remove unused high-cardinality columns (GUIDs, millisecond timestamps); each one disproportionately inflates VertiPaq's size.
- **Split Date and Time:** instead of a single DateTime column, create a Date column (linked to the calendar table) and a separate Time column if you need the time portion.
- **Disable “Auto Date/Time”:** File > Options > Data Load > Time intelligence. Prevents Power BI from creating a hidden calendar table for every date column in the model.
- **Strict star schema:** facts hold metrics and foreign keys; dimensions hold descriptive attributes; relationships are 1:N with a single filter direction.
- **Always use explicit measures:** never rely on implicit aggregation when dragging a numeric field; ALWAYS create a named measure such as Total Sales = SUM(...).
- **Avoid FILTER over entire tables:** prefer CALCULATE(..., Table[Column] = "Value") or KEEPFILTERS over FILTER(Table, Table[Column] = "Value") whenever possible.
- **Use variables (VAR):** avoid recalculating the same sub-expression multiple times within a measure; variables also make DAX more readable and easier to debug.
- **Measure before optimizing:** use Performance Analyzer (visual by visual) and DAX Studio (query plan, Formula Engine vs. Storage Engine) to locate the real bottleneck before rewriting anything.

## 6.4 Load strategy by data volume (quick reference)

| Volume / need                                      | Recommended strategy                                                                                         | Trade-off to accept                                                              |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| 5-100M+ rows, no real-time need                    | Import + incremental refresh (RangeStart/RangeEnd)                                                           | Requires configuring date-based partition parameters                             |
| Data that must be “second-fresh”                   | DirectQuery                                                                                                  | Every user interaction fires SQL at the source; requires very well-tuned indexes |
| Massive fact tables (hundreds of millions of rows) | Composite model with aggregations (aggregate table in Import + detail in DirectQuery via Dual relationships) | Requires defining aggregation mappings and Dual relationships on dimensions      |

## 7. Exam-day strategy

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- 100 minutes, ~40-60 questions, including at least one case study with several questions tied to the same scenario — read it carefully the first time, don't skip ahead.
- Time management: if a question blocks you for more than 90 seconds, flag it for review and move on; come back at the end.
- “Drag and order the steps” questions usually show up around Power Query and RLS/gateway setup: practice the logical order, not just the end result.
- When torn between two DAX options, ask first: “does this change the filter context or not?” — that resolves most doubts between CALCULATE, FILTER, and simple aggregations.
- The day before, only review the cheat sheet (no new material) and get a good night's sleep: the exam measures pattern recognition, not rote memorization.

## 8. Recommended resources

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- Microsoft Learn — the official “Power BI Data Analyst” learning path (free, mapped 1:1 to the syllabus).
- Microsoft's DAX documentation ([learn.microsoft.com/dax](https://learn.microsoft.com/dax)) and SQLBI.com for reference articles on row/filter context.
- DAX Studio and Tabular Editor (both free) for hands-on optimization practice.
- Microsoft's official practice exams (MeasureUp) to simulate real conditions.
- Power BI Desktop itself, using sample datasets (Contoso, AdventureWorks) to replicate every exercise in this plan.

## 9. Complete catalog: visualizations, filters and Power BI functionality

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This section is intentionally exhaustive. Many of these concepts follow patterns common across BI tools, but the exact names, behavior, and menu locations are specific to Power BI — and that is exactly how the exam tests them.

### 9.1 Exhaustive catalog of visualization types

Power BI Desktop's Visualizations pane ships with native visuals, and lets you import hundreds more from AppSource (the marketplace) or build your own with the developer SDK. For exam purposes, group them like this:

#### A. Comparison and trend

- Column chart: clustered, stacked, and 100% stacked (vertical).
- Bar chart: clustered, stacked, and 100% stacked (horizontal).

- Line chart (including lines with markers, and continuous or categorical date axes).
- Area chart and stacked area chart.
- Combo chart (columns and lines) with one or two Y axes.
- Waterfall chart — for breaking down incremental changes to a total.
- Funnel chart — for sequential processes with drop-off in volume (e.g. a sales pipeline).
- Ribbon chart — for tracking rank changes between categories over time.

## **B. Distribution and relationship**

- Scatter chart and bubble chart, including the play axis to animate over time.
- Histogram (built by binning a continuous measure).
- Forecasting lines overlaid on a time series, with confidence bands.
- Reference lines and error bands (constant, average, percentile, min/max).

## **C. Composition**

- Pie chart.
- Donut chart.
- Treemap (hierarchy shown through size and color).

## **D. Tables and matrices**

- Table — flat rows, no hierarchy.
- Matrix — hierarchical rows/columns, expand/collapse, configurable totals/subtotals; supports drill-down by level.

## **E. Cards and KPIs**

- Card — a single highlighted value.
- Multi-row card — several metrics per category in card format.
- KPI visual — current value vs. target with a trend indicator.
- Gauge — a value within a range with a marked target.

## **F. Maps**

- Map (Bing-based) — geolocated bubbles.
- Filled map / choropleth — regions colored by value.
- Shape map — using a custom TopoJSON file (e.g. states, districts).
- ArcGIS Maps for Power BI and Azure Maps — advanced mapping visuals available via AppSource.

## **G. Analytical, hierarchical and AI visuals**

- Decomposition tree — interactive breakdown of a measure across multiple dimensions, with AI-suggested splits (“High value”/“Low value”).

- Key influencers — identifies which variables most influence a categorical or continuous outcome.
- Q&A visual — natural-language query embedded directly as a visual.
- Smart narrative / Copilot-generated narrative — automatically generates descriptive text about the report's data.
- Clustering — automatic grouping available as an analysis option on scatter charts.

## H. Slicers (as a visual type)

- Vertical or horizontal list (with or without a search box).
- Dropdown.
- Numeric slider (between / greater than / less than).
- Date slicer with a relative-date selector (last week, month, quarter...).
- Hierarchical slicer (several fields across levels, e.g. Year > Quarter > Month).

## I. Other canvas elements

- Button — with actions: page navigation, bookmark, drillthrough, back, refresh page, web URL.
- Image, text box, and shape — static design elements.
- Web content — embeds an external page or widget.
- Custom R and Python visuals (R script visual / Python script visual).
- Visuals imported from AppSource (marketplace) and internally certified organizational visuals.
- Metrics/Scorecard visual — for tracking goals (OKRs) linked to measures in the model.

## 9.2 All filter types

Power BI applies filters in a cascade based on level: report first, then page, then visual, and finally drillthrough — each level can further restrict, but never expand, what the previous level allows.

| Level               | Scope                                                         | Example                                                            |
|---------------------|---------------------------------------------------------------|--------------------------------------------------------------------|
| Report-level filter | Affects every page in the report                              | Excluding the current (incomplete) year across the whole report    |
| Page-level filter   | Affects only the current page                                 | Showing only the “Europe” region on the detail page                |
| Visual-level filter | Affects only that specific visual                             | Top 10 products in a table, without affecting the rest of the page |
| Drillthrough filter | Applied automatically when navigating to the destination page | Filtering by the specific customer that was clicked                |

## Filter condition types (within any level)

- Basic filter: selecting values from a list, with a search box.
- **Advanced filter:** logical conditions (is, is not, contains, starts with, is blank...) combinable with AND/OR — up to two conditions in the standard panel.

- **Top N:** the N elements with the highest/lowest value of a measure, optionally combined with a basic filter.
- **Relative date filtering:** e.g. “last 30 days”, “current quarter” — recalculated fresh every time the report is opened.
- **Relative time filtering:** the same idea, but at an hours/minutes granularity, typical of near-real-time data.
- **Filter on a measure/aggregation:** e.g. “show only categories whose total sales exceed 10,000”.

### Interactive mechanisms related to filtering

- Cross-highlighting vs. cross-filtering: clicking one visual can either highlight the corresponding portion of another visual or filter it completely — configured under “Edit interactions”.
- Include/Exclude (right-click on a data point) — creates a one-off filter without touching the Filters pane.
- Single vs. multiple selection on slicers and visuals (Ctrl+click for multi-select).
- Locking filters and hiding the Filters pane from the end user of a published report.
- Sync Slicers across several pages of the report.
- What-if parameters acting as a dynamic filter driven by a numeric slicer.
- Field parameters — let the end user swap which dimension or measure feeds an axis/visual, effectively acting as a “structural filter”.
- Drill down / Drill up / “Expand to next level” — change the visible granularity; not filters in the strict sense, but they share menus and are often confused with filters on the exam.
- **Conceptual note:** Row-Level Security (RLS) also “filters” data, but it is a security mechanism fixed per role, not an interactive filter — it never appears in the end user's Filters pane.

## 9.3 Additional functionality across the Power BI ecosystem

Pieces of the ecosystem the syllabus references indirectly (“Copilot”, “distribution”, “content promotion”) that are worth recognizing by name and function:

| Feature                 | What it is                                                                                 |
|-------------------------|--------------------------------------------------------------------------------------------|
| Q&A                     | Query the model by typing a natural-language question; a visual is generated automatically |
| Quick Insights          | AI that analyzes a table and automatically surfaces patterns or findings                   |
| What-if parameters      | A numeric slicer that feeds a DAX measure to simulate scenarios                            |
| Field parameters        | A parameter that lets you swap fields/measures within the same visual                      |
| Metrics / Scorecards    | Goal tracking (OKRs) linked to measures, with checkpoints and history                      |
| Paginated reports       | Fixed-layout, invoice/PDF-style reports built in Power BI Report Builder                   |
| Dataflows (Gen1 / Gen2) | Reusable Power Query logic shared across several models, executed in the Service           |

| Feature                             | What it is                                                                                                                  |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Datamarts                           | A managed analytical database inside the Service, with no external infrastructure, including its own model and SQL querying |
| Template apps                       | Prebuilt, parameterizable report solutions distributed to customers or SaaS data sources                                    |
| Power BI Mobile                     | iOS/Android/Windows apps with mobile-optimized layouts and data alerts                                                      |
| Power BI Embedded                   | Embedding reports into your own or a third party's applications (for customers or for your own organization)                |
| Data alerts and subscriptions       | Automatic notifications when a card/KPI crosses a threshold, or scheduled email delivery                                    |
| Copilot                             | Natural-language chat, page generation, narrative writing, and semantic model summarization                                 |
| Admin portal / tenant settings      | Organization-wide governance: which features are enabled, who can share externally, etc.                                    |
| Sensitivity labels                  | Integration with Microsoft Purview to classify and protect reports based on data sensitivity                                |
| Content certification and promotion | "Promoted" (any user) vs. "Certified" (admins only) badges to flag trustworthy content                                      |
| Analyze in Excel / Export           | Consuming the semantic model as an Excel PivotTable, or exporting to PDF/PPT/Excel/.pbix                                    |
| Teams and SharePoint integration    | Embedding Power BI tabs directly inside Teams or SharePoint pages                                                           |
| Presentation mode and Focus mode    | Expanding a single visual to full screen within the report itself                                                           |
| Personalize visuals                 | Lets the end user of a report build their own variant of a visual without editing the original report                       |

### How to study this section

Don't try to memorize the table in one pass: open Power BI Desktop, drop each visual onto the canvas once, and notice which wells it asks for in the Visualizations pane. That muscle memory of "where everything lives" is what the exam is really testing.

## 10. Final checklist before you sit the exam

- ☐ I can explain the difference between Import, DirectQuery, Dual and DirectLake without hesitating
- ☐ I know which Power Query steps break query folding
- ☐ I clearly distinguish Merge (JOIN) from Append (UNION)
- ☐ I can design a star schema from a denormalized table
- ☐ I can explain row context vs. filter context with my own example
- ☐ I can write a CALCULATE measure with a filter modifier (ALL/ALLEXCEPT/KEEPFILTERS) from memory
- ☐ I can implement time intelligence (SAMEPERIODLASTYEAR, DATESYTD)
- ☐ I know when to use a calculated column vs. a measure
- ☐ I can configure both a static and a dynamic RLS role
- ☐ I know when a gateway is needed and which type
- ☐ I know the 4 workspace roles and what each one can do

- ☐ I can configure a drillthrough and a bookmark with a button
- ☐ I can use Performance Analyzer to find a slow visual
- ☐ I recognize the ~25 native visual types and know when to use each one
- ☐ I can distinguish the 4 filter levels (report/page/visual/drillthrough) and their order of application
- ☐ I can differentiate basic, advanced, Top N, relative-date and relative-time filters
- ☐ I can explain the difference between cross-filtering and cross-highlighting
- ☐ I can recognize by name: Q&A, Quick Insights, dataflows, datamarts, template apps and paginated reports
- ☐ I know what a field parameter and a What-if parameter are, and how they differ